SUHYUN CHOI

Integrated M.S./Ph.D. Student · Reinforcement Learning & Sequential Decision Making
Hanyang University · Information and Intelligence Systems Lab (IISL) · Seoul, Republic of Korea
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RESEARCH PROFILE

I develop reinforcement learning and sequential decision-making methods for reliable policy learning from imperfect and limited data, with a focus on offline and offline-to-online RL, goal-conditioned long-horizon decision making, and optimization-based policy improvement.

EDUCATION

Hanyang University 2024–Present
Integrated M.S./Ph.D. Program in Electronic Engineering
Information and Intelligence Systems Lab (IISL) · Advisor: Prof. Songnam Hong

MANUSCRIPT

S. Choi, S. Cho, and S. Hong, "PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning."
Under Review, AAAI 2027.

RESEARCH EXPERIENCE

PathBridger — Offline Goal-Conditioned Reinforcement Learning 2026
- Developed a hierarchical offline GCRL approach that constructs explicit state-space bridges, decodes them into short executable action chunks via inverse dynamics, and replans during execution.
- Evaluated long-horizon navigation and manipulation behavior on OGBench tasks, including settings with limited offline data.

Calibrated Adaptive Multi-step Proximal Improvement (CAPO) 2026–Present
- Investigating calibrated, bounded policy refinement in offline RL when critic estimates are imperfect and distribution shift grows away from dataset support.
- Developing principled criteria for reliable policy improvement that account for value-estimation uncertainty and optimization error.

SK hynix–Hanyang University Industry–Academic Research Project 2024–2025
- Contributed to machine-learning- and transfer-function-based modeling for circuit-interconnect waveform and delay estimation, including model validation and industry technical exchanges.

HONORS & FUNDING

- Outstanding Graduation Project Award — KRW 2,000,000 award
- Excellence Award, College of Engineering Capstone Design — KRW 1,000,000 award
- BK Research Assistantship (BK-RA), Hanyang University — 2025

SELECTED GRADUATE COURSEWORK

Learning & Control: Machine Learning; Robot Learning; Advanced Optimal Control Theory
Probability & Analysis: Probability and Statistics; Stochastic Processes; Uncertainty Quantification; Functional Analysis I